STUDENT PERFORMANCE PREDICTION

Machine Learning Project Presentation

Prepared By:

vignesh S

Ajay Krishnan

Introduction

Student Performance Prediction using Machine Learning
Goal: Predict student exam marks using study related factors helps teachers to understand student performance early.

PROBLEM STATEMENT

Students have different habits and attendance levels

Predict exam marks based on learning behaviour

Build a model that estimates performance before exams.

TOOLS AND TECHNOLOGIES

Python Programming Language

Pandas and NumPy for data Processing

Scikit-learn for Machine Learning

Streamlit for browser interface

DATASET

Generated dataset for 1000 students

Features: Study Hours, Sleep Hours, Attendance

Target variable: Students Marks

DATA PROCESSING

Dataset converted into structured dataframe

Input features (X) and output variable (Y) separated

Dataset split into training and testing sets

MACHINE LEARNING MODEL

Linear Regression algorithm used

Model trained using training dataset

Learns relationship between study behaviour and marks

MODEL ACCURACY

Model tested using testing dataset

R2 score used for accuracy measurement

High score means better prediction performance

WEB APPLICATION

Interface developed using Streamlit

User inputs study hours, sleep hours and attendance

System predicts expected marks

CONCLUSION

Machine Learning helps predict academic performance

Useful for early performance analysis

Can be improved using real world student datasets